

Clinical Update

ADAPTED FROM:

2025 ACC/AHA/ACEP/NAEMSP/SCAI Guideline for
the Management of Patients with Acute Coronary
Syndromes.

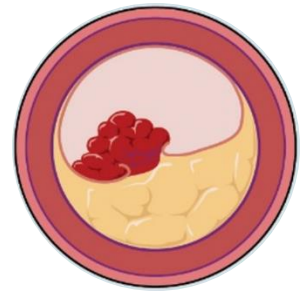


Table 1.

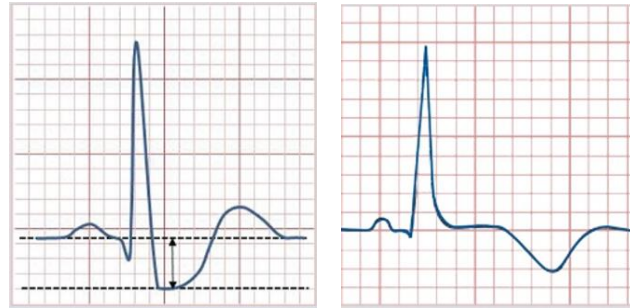
Applying Class of Recommendation and Level of Evidence to Clinical Strategies, Interventions, Treatments, or Diagnostic Testing in Patient Care

CLASS (STRENGTH) OF RECOMMENDATION	LEVEL (QUALITY) OF EVIDENCE‡
CLASS 1 (STRONG) Benefit >>> Risk Suggested phrases for writing recommendations: • Is recommended • Is indicated/useful/effective/beneficial • Should be performed/administered/other • Comparative-Effectiveness Phrases†: – Treatment/strategy A is recommended/indicated in preference to treatment B – Treatment A should be chosen over treatment B	LEVEL A • High-quality evidence‡ from more than 1 RCT • Meta-analyses of high-quality RCTs • One or more RCTs corroborated by high-quality registry studies
	LEVEL B-R (Randomized) • Moderate-quality evidence‡ from 1 or more RCTs • Meta-analyses of moderate-quality RCTs
CLASS 2a (MODERATE) Benefit >> Risk Suggested phrases for writing recommendations: • Is reasonable • Can be useful/effective/beneficial • Comparative-Effectiveness Phrases†: – Treatment/strategy A is probably recommended/indicated in preference to treatment B – It is reasonable to choose treatment A over treatment B	LEVEL B-NR (Nonrandomized) • Moderate-quality evidence‡ from 1 or more well-designed, well-executed nonrandomized studies, observational studies, or registry studies • Meta-analyses of such studies
	LEVEL C-LD (Limited Data) • Randomized or nonrandomized observational or registry studies with limitations of design or execution • Meta-analyses of such studies • Physiological or mechanistic studies in human subjects
CLASS 2b (Weak) Benefit ≥ Risk Suggested phrases for writing recommendations: • May/might be reasonable • May/might be considered • Usefulness/effectiveness is unknown/unclear/uncertain or not well-established	LEVEL C-EO (Expert Opinion) • Consensus of expert opinion based on clinical experience.
	COR and LOE are determined independently (any COR may be paired with any LOE). A recommendation with LOE C does not imply that the recommendation is weak. Many important clinical questions addressed in guidelines do not lend themselves to clinical trials. Although RCTs are unavailable, there may be a very clear clinical consensus that a particular test or therapy is useful or effective. * The outcome or result of the intervention should be specified (an improved clinical outcome or increased diagnostic accuracy or incremental prognostic information). † For comparative-effectiveness recommendation (COR 1 and 2a; LOE A and B only), studies that support the use of comparator verbs should involve direct comparisons of the treatments or strategies being evaluated. ‡ The method of assessing quality is evolving, including the application of standardized, widely-used, and preferably validated evidence grading tools; and for systematic reviews, the incorporation of an Evidence Review Committee. COR indicates Class of Recommendation; EO, expert opinion; LD, limited data; LOE, Level of Evidence; NR, nonrandomized; R, randomized; and RCT, randomized controlled trial.
CLASS 3: No Benefit (MODERATE) Benefit = Risk Suggested phrases for writing recommendations: • Is not recommended • Is not indicated/useful/effective/beneficial • Should not be performed/administered/other	
CLASS 3: Harm (STRONG) Risk > Benefit Suggested phrases for writing recommendations: • Potentially harmful • Causes harm • Associated with excess morbidity/mortality • Should not be performed/administered/other	

Definition and Classifications of Acute Coronary Syndromes



Non-Occlusive Thrombus



ST Depression or T Wave Inversion
(May be electrically silent)

Biomarker Negative

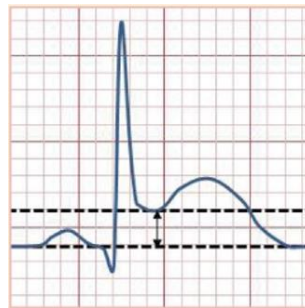
Unstable Angina

Biomarker Positive

NSTEMI



Occlusive Thrombus



ST Elevation

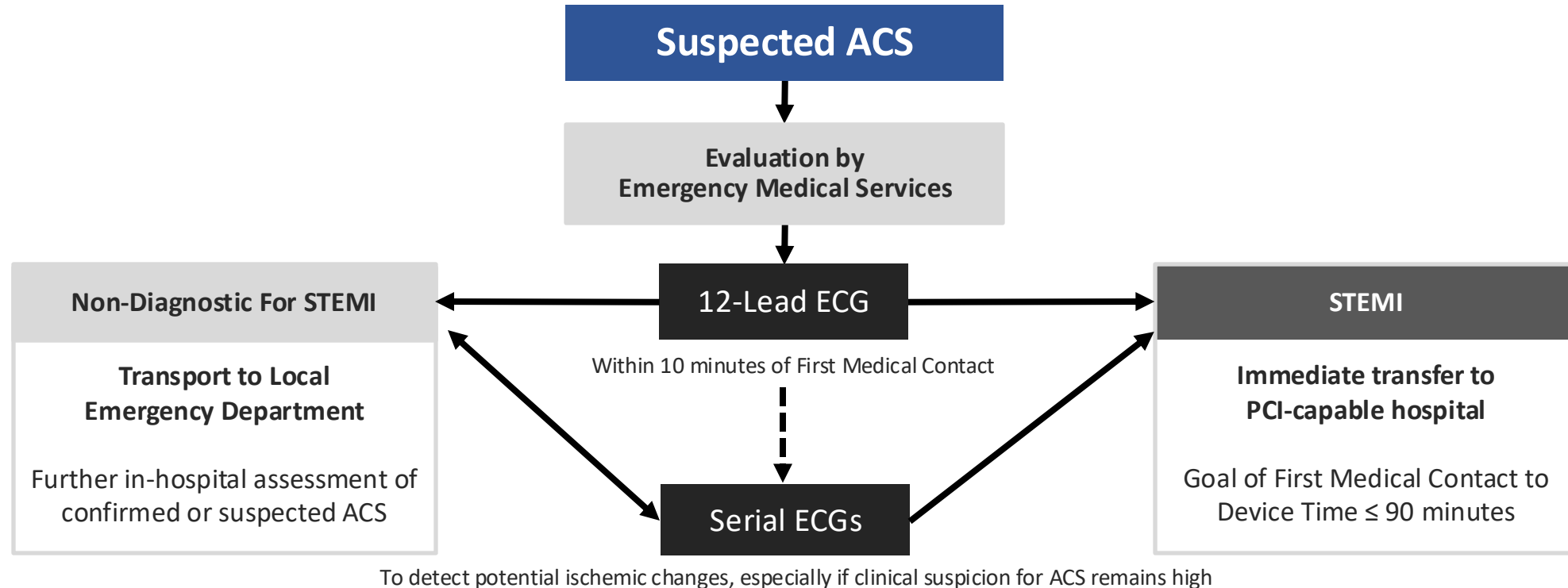
Biomarker Positive

STEMI

(May be negative if drawn too early from symptom onset)

Abbreviations: NSTEMI indicates non-ST-elevation myocardial infarction; and STEMI, ST-elevation myocardial infarction.

Pre-hospital Assessment and Management Considerations for Suspected ACS



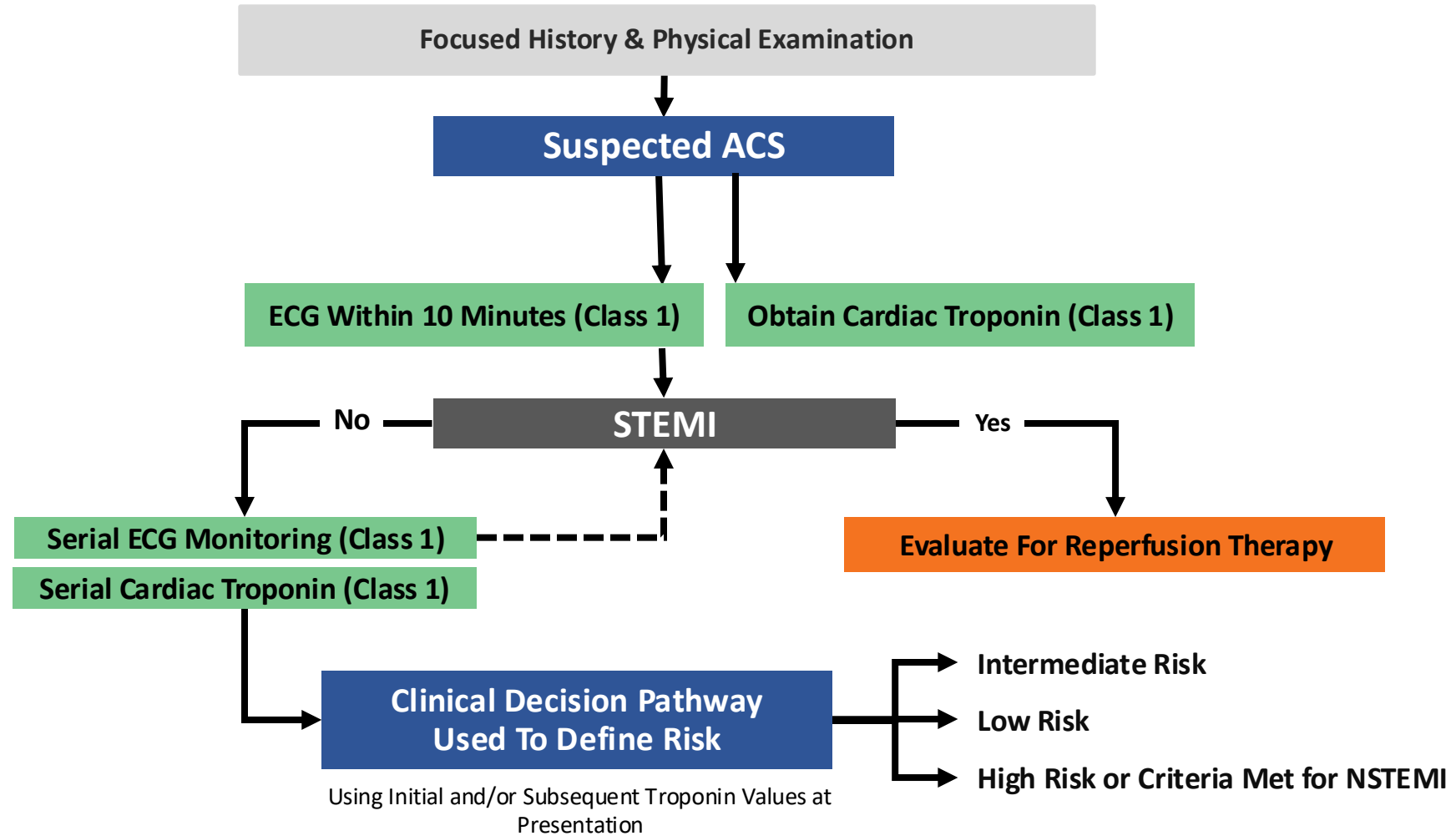
In patients with STEMI managed with primary PCI

↑ each 30 minute delay is associated with ↑ 7.5% relative risk of 1-year death

Abbreviations: ACS indicates acute coronary syndrome; ECG, electrocardiogram; STEMI, ST-elevation myocardial infarction; and PCI, percutaneous coronary intervention.

Rao, S.V., et al. 2025 AHA/ACC/ACEP/NAEMSP/SCAI Guideline for Acute Coronary Syndromes. *Circulation*.

Initial **In-Hospital** Assessment of Patients with Confirmed or Suspected ACS

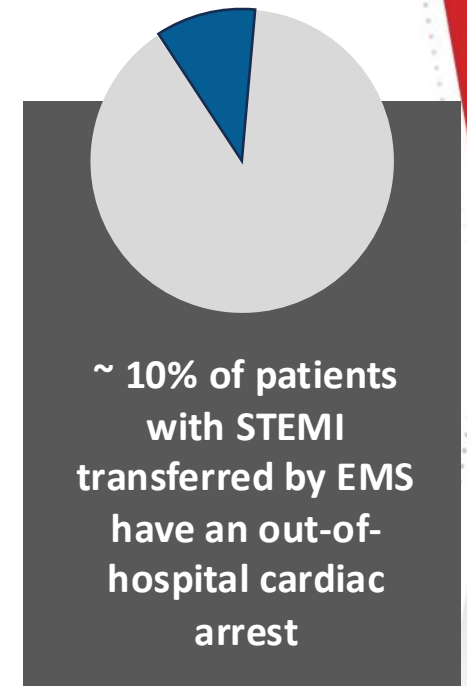


Abbreviations: ACS indicates acute coronary syndrome; ECG, electrocardiogram; and STEMI, ST-elevation myocardial infarction.

Management of Patients Presenting with Cardiac Arrest

Following achievement of return of spontaneous circulation (ROSC)

Mental Status	Awake	Comatose	Comatose	Comatose
Presence of STEMI	Yes	Yes	Yes	No
Prognostic Features	--	Favorable	Unfavorable	--
Guideline Recommendation	PPCI (Class 1)	PPCI (Class 1)	PPCI Reasonable After Individualized Assessment (Class 2b)	Immediate Coronary Angiography Not Recommended (Class 3: No Benefit)



Abbreviations: EMS indicates emergency medical services; PPCI, primary percutaneous coronary intervention; and STEMI, ST-elevation myocardial infarction.

Standard Medical Therapy for Acute Coronary Syndromes

Analgesic Treatment Options

Medication	Route	Considerations
Nitroglycerin (SL)	0.4 mcg sublingual every 5 minutes for up to 3 doses	Avoid use in suspected RV infarction or SBP < 90 mm Hg
Nitroglycerin (IV)	Start at 10 mcg/min and titrate to pain relief and hemodynamic tolerability	- Consider for persistent anginal pain after oral nitrate therapy - Use if ACS is complicated by hypertension or flash pulmonary edema - Avoid use in suspected RV infarction or SBP < 90 mm Hg
Morphine (IV)	2-4 mg; may repeat if needed every 5-15 minutes up to 10 mg total dose	- Use for pain that is resistant to maximal anti-ischemic medications - May delay the effects of oral P2Y12 therapy
Fentanyl (IV)	25-50 mcg; may repeat if needed up to 100 mcg total dose	- Use for pain that is resistant to maximally tolerated anti-ischemic medications - May delay the effects of oral P2Y12 therapy

Analgesic therapies provide symptomatic relief but have not been shown to improve clinical outcomes in ACS. If ischemic symptoms persist despite efforts at pain control, consider urgent coronary angiography.

Antiplatelet Therapy: Aspirin During Hospitalization



Aspirin	
COR	RECOMMENDATIONS
1	 In patients with ACS , an initial oral loading dose of aspirin , followed by daily low-dose aspirin is recommended to reduce death and MACE

Abbreviations: ACS indicates acute coronary syndrome; and MACE, major adverse cardiovascular event.

Rao, S.V., et al. 2025 AHA/ACC/ACEP/NAEMSP/SCAI Guideline for Acute Coronary Syndromes. *Circulation*.

Antiplatelet Therapy: Oral P2Y12 Inhibitors During Hospitalization



P2Y12 Inhibitors

COR	RECOMMENDATIONS
1	 In patients with ACS, an oral P2Y12 inhibitor should be administered in addition to aspirin to reduce MACE
3: HAR M	 In patients with a history of stroke or TIA, prasugrel should NOT be administered because of worse net clinical outcomes

Oral P2Y12 Inhibitors: In-Hospital Management of Patients with NSTEMI-ACS



In patients with NSTEMI-ACS undergoing PCI, prasugrel or ticagrelor is recommended to reduce MACE and stent thrombosis.
(Class 1)



In patients with NSTEMI-ACS who are managed without planned invasive evaluation, ticagrelor is recommended to reduce MACE.
(Class 1)



In patients with NSTEMI-ACS, clopidogrel is recommended to reduce MACE when prasugrel or ticagrelor are not available, cannot be tolerated, or are contraindicated.
(Class 1)



In patients with NSTEMI-ACS planned for an invasive strategy with timing of angiography anticipated to be >24h, upstream treatment with clopidogrel or ticagrelor may be considered to reduce MACE.
(Class 2b)

Oral P2Y12 Inhibitors: In-Hospital Management of Patients with STEMI



In patients with STEMI managed with PPCI, prasugrel or ticagrelor should be administered to reduce MACE and stent thrombosis.
(Class 1)



In patients with STEMI managed with PPCI, clopidogrel is recommended to reduce MACE and stent thrombosis when prasugrel or ticagrelor are not available, cannot be tolerated, or are contraindicated.
(Class 1)



In patients with STEMI managed with fibrinolytic therapy, clopidogrel should be administered concurrently to reduce death and MACE.
(Class 1)

Antiplatelet Therapy: Intravenous P2Y12

Intravenous P2Y12 Inhibitors

COR	RECOMMENDATIONS
2b	 Among patients with ACS undergoing PCI who have not received a P2Y12 inhibitor, intravenous cangrelor may be reasonable to reduce periprocedural ischemic events

Intravenous Cangrelor: Rapid and potent platelet inhibitory effects with restoration of platelet function occurring within one hour of drug discontinuation

Consider in clinical scenarios where:



Absorption of orally administered P2Y12 inhibitors is impaired or not possible



Patients requiring CABG or other surgery early after PCI when prolonged discontinuation of a P2Y12 inhibitor is not thought to be safe

The transition from intravenous to oral P2Y12 inhibition is an important consideration to ensure adequate platelet inhibition upon completion of cangrelor infusion

Antiplatelet Therapy: Intravenous Glycoprotein IIb/IIIa Inhibitors

Intravenous Glycoprotein IIb/IIIa Inhibitors	
COR	RECOMMENDATIONS
2a	 In patients with ACS undergoing PCI with large thrombus burden, no-reflow, or slow flow, adjunctive use of an intravenous or intracoronary glycoprotein IIb/IIIa inhibitor is reasonable to improve procedural success and reduce infarct size
3: HARM	 In patients with ACS, glycoprotein IIb/IIIa inhibitors should <u>not be administered</u> routinely due to lack of ischemic benefit and increased risk of bleeding

Abbreviations: ACS indicates acute coronary syndrome; and PCI, percutaneous coronary intervention.

Parenteral Anticoagulation

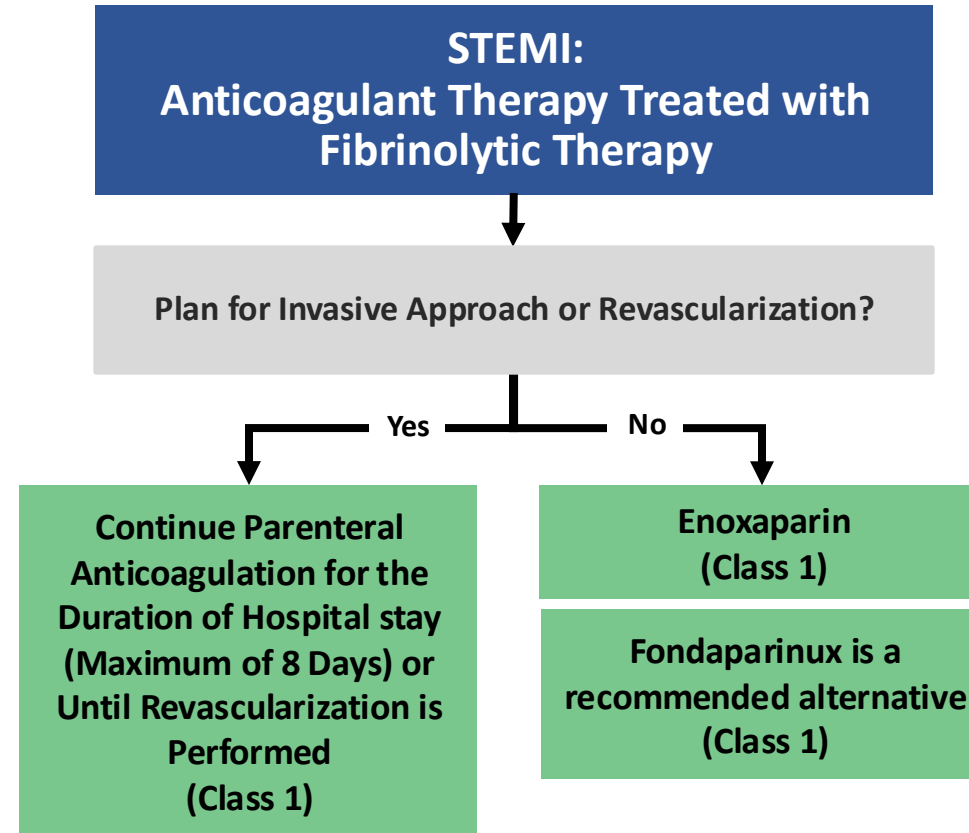
PCI Not Planned	
COR	RECOMMENDATIONS
1	In patients with <u>NSTE-ACS</u> , intravenous unfractionated heparin (UFH) is useful to reduce ischemic events.
1	In patients with <u>NSTE-ACS</u> in whom an <u>early invasive approach</u> is <u>not anticipated</u> , either enoxaparin or fondaparinux are recommended alternatives to UFH.

Coronary Revascularization	
COR	RECOMMENDATIONS
1	In patients with ACS undergoing coronary revascularization (CABG or PCI) in the same admission, parenteral anticoagulation should be continued until revascularization to reduce ischemic events.

PCI Planned	
COR	RECOMMENDATIONS
1	In patients with <u>ACS</u> undergoing <u>PCI</u> , intravenous UFH is useful to reduce ischemic events.
1	In patients with <u>STEMI</u> undergoing <u>PCI</u> , bivalirudin is useful as an alternative to UFH to reduce mortality and bleeding
2b	In patients with <u>NSTE-ACS</u> undergoing <u>PCI</u> , bivalirudin may be reasonable as an alternative to UFH to reduce bleeding
2b	In patients with <u>ACS</u> , intravenous enoxaparin may be considered as an alternative to UFH at the time of PCI to reduce ischemic events
3: HARM	In patients with ACS, <u>fondaparinux should NOT be used to support PCI</u> because of the risk of catheter thrombosis

Abbreviations: ACS indicates acute coronary syndrome; NSTE, non-ST elevation; PCI, percutaneous coronary intervention; STEMI, ST elevation myocardial infarction; and UFH, unfractionated heparin.

Parenteral Anticoagulation



Lipid Management

Statin Therapy	
COR	RECOMMENDATIONS
1	 In patients with ACS, high-intensity statin therapy is recommended to reduce the risk of MACE
1	 In patients with ACS who are already on maximally tolerated statin therapy with LDL ≥ 70 mg/dL (≥ 1.8 mmol/L) , adding a non-statin lipid lowering agent is recommended to further reduce the risk of MACE
2a	 In patients with ACS who are already on maximally tolerated statin therapy with LDL 55-69 mg/dL (≥ 1.4 - < 1.8 mmol/L) , adding a non-statin lipid lowering agent is reasonable to reduce the risk of MACE
2b	 In patients with ACS, the concurrent initiation of ezetimibe in combination with maximally tolerated statin may be considered to reduce the risk of MACE

Statin Intolerance	
COR	RECOMMENDATIONS
1	 In patients with ACS who are <u>statin intolerant</u> , non-statin lipid lowering therapy is recommended to lower LDL and reduce the risk of MACE

Non-Statin Lipid Lowering Therapies:



Ezetimibe



PCSK9 Inhibitors
(monoclonal antibodies or inclisiran)



Bempedoic Acid

Beta Blocker Therapy and Renin-Angiotensin System Inhibitors

All patients without contraindication**



Early (<24 h) initiation of oral beta blocker therapy to reduce risk of reinfarction and ventricular arrhythmias (Class 1)

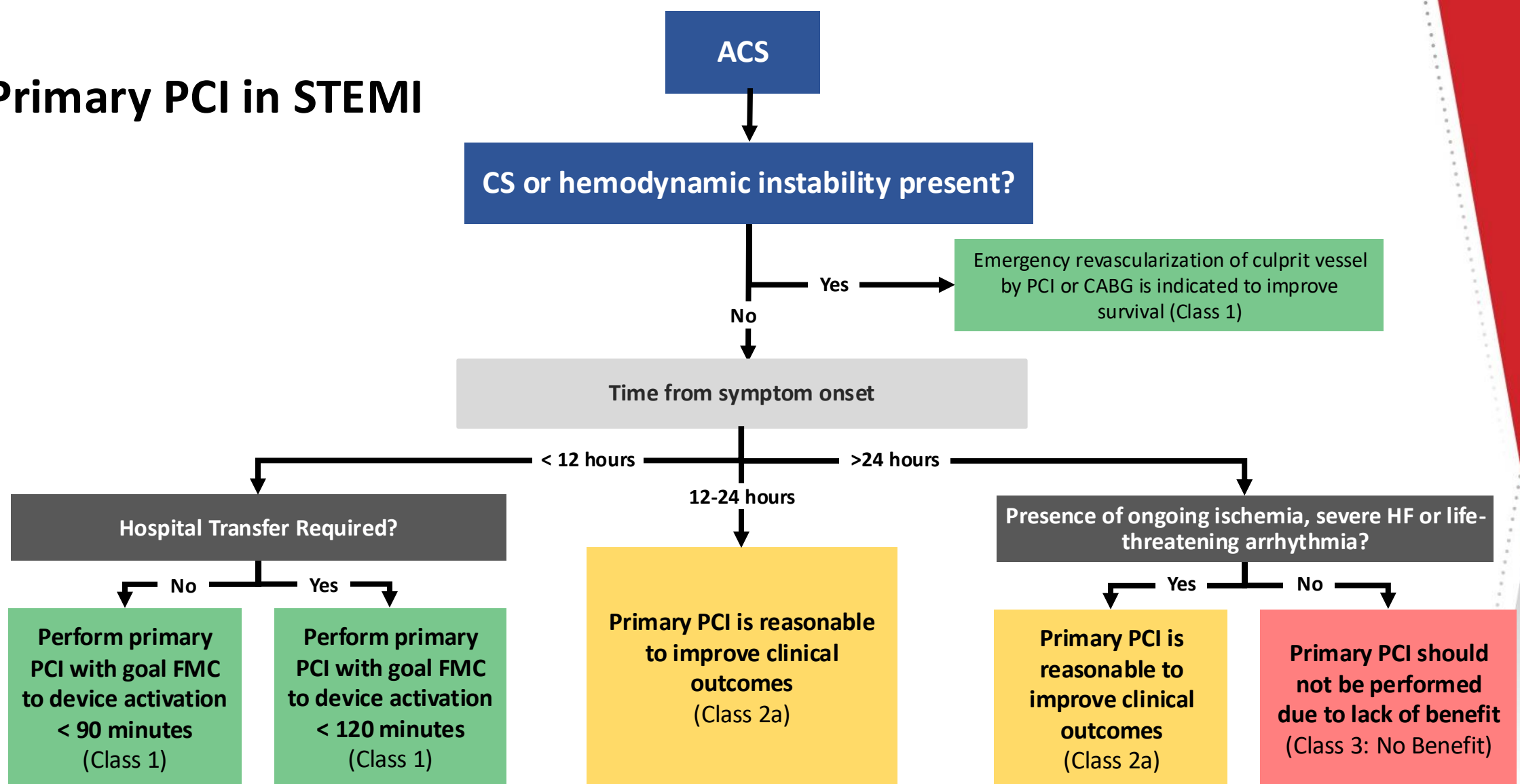
**Contraindications to Beta Blocker Therapy

- Acute HF
- Low output state or risk for CS
- PR > 0.24 ms
- 2nd or 3rd degree AVB without a pacemaker
- Severe bradycardia
- Active bronchospasm

COR	RECOMMENDATIONS
1	In high-risk patients with ACS (LVEF ≤40%, HTN, diabetes mellitus or STEMI with anterior location), an oral ACEi or an ARB is indicated to reduce all-cause mortality and MACE.
1	In patients with ACS and LVEF ≤ 40%, and with HF symptoms and/or diabetes mellitus, a MRA is indicated to reduce all-cause mortality and MACE.
2a	In ACS patients who are not considered high risk, an oral ACEi or an ARB is reasonable to reduce MACE.

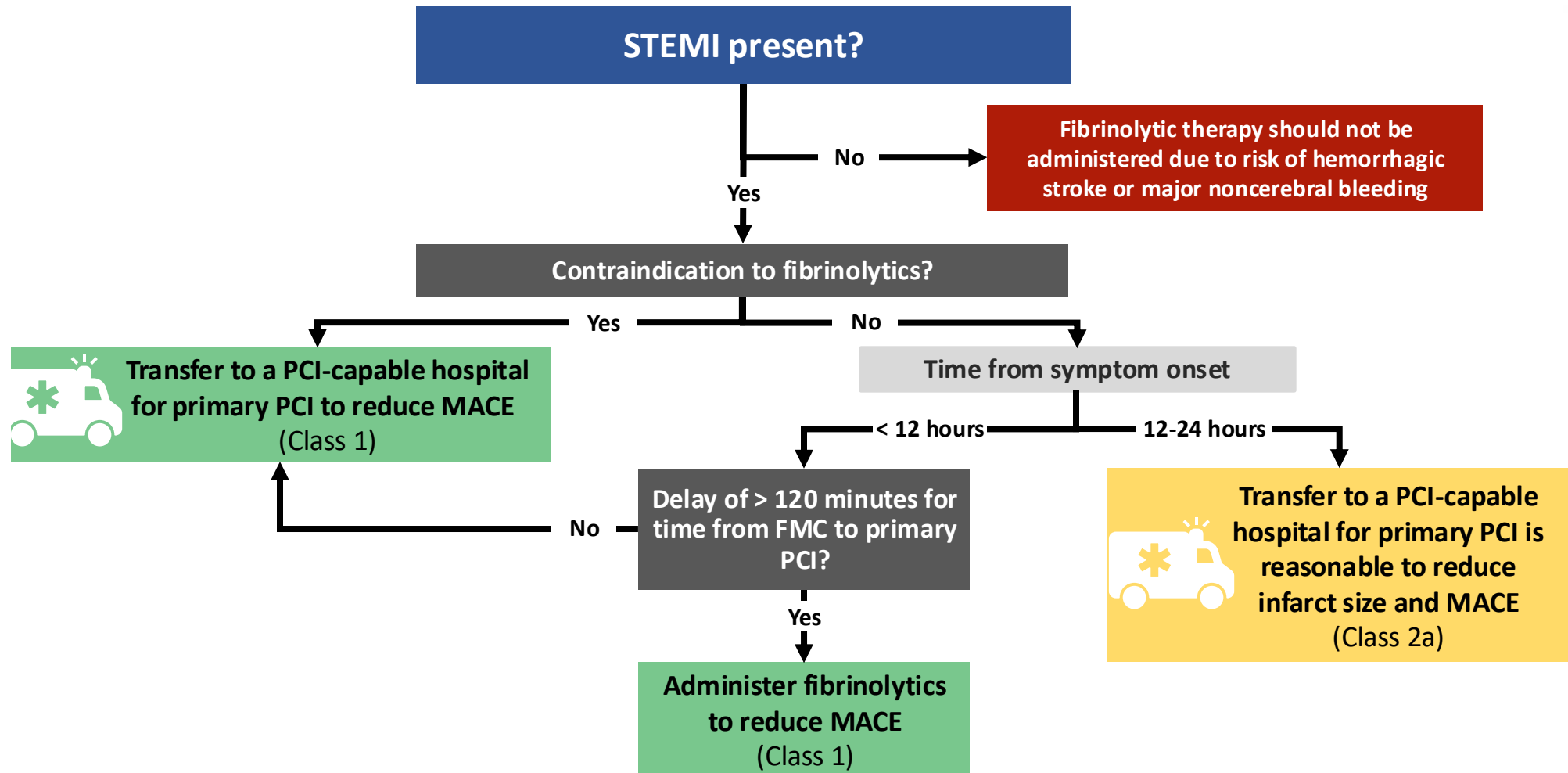
Abbreviations: ACEi indicates angiotensin-converting enzyme inhibitor; ACS, acute coronary syndrome; ARB, angiotensin receptor blocker; CS, cardiogenic shock; HF, heart failure; HTN, hypertension; LVEF, left ventricular ejection fraction; MACE, major adverse cardiovascular event; MRA, mineralocorticoid receptor antagonist; and STEMI, ST-elevation myocardial infarction.

Primary PCI in STEMI



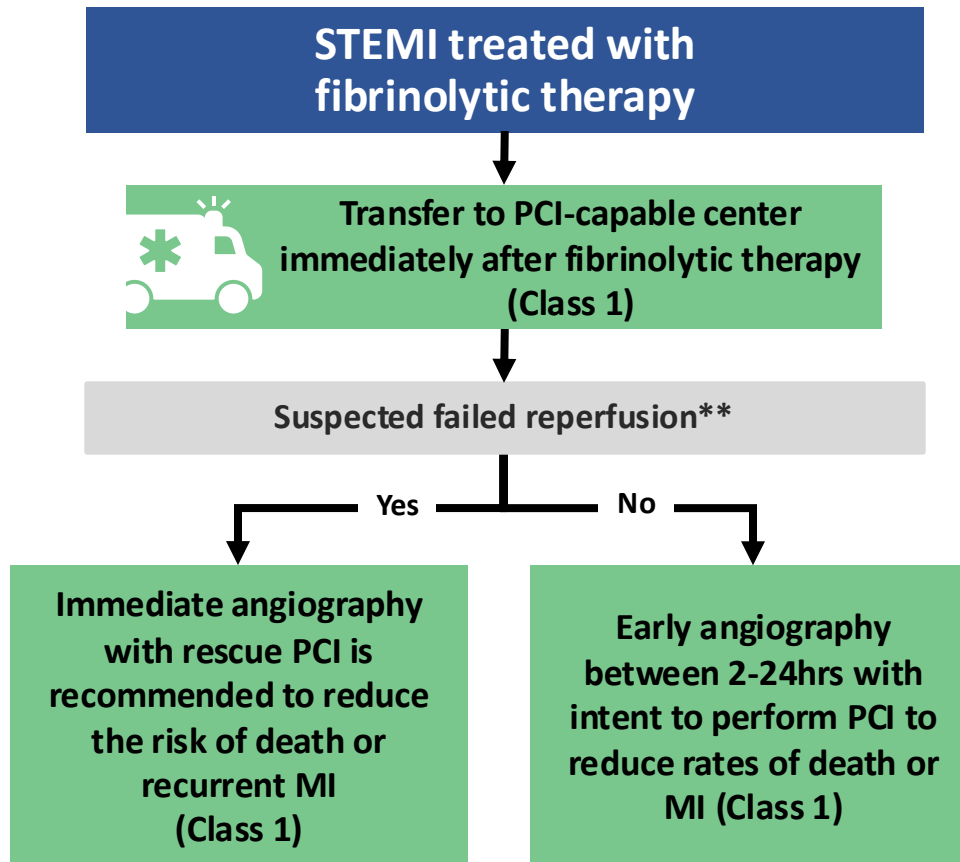
Abbreviations: ACS indicates acute coronary syndrome; CS, cardiogenic shock; CABG, coronary artery bypass graft; FMC, first medical contact; HF, heart failure; STEMI, ST-elevation myocardial infarction; and PCI, percutaneous coronary intervention.

Reperfusion at Non-PCI Capable Hospitals



Abbreviations: FMC indicates first medical contact; MACE, major adverse cardiovascular event; STEMI, ST-elevation myocardial infarction; and PCI, percutaneous coronary intervention.

Coronary Angiography and PCI After Fibrinolytic Therapy

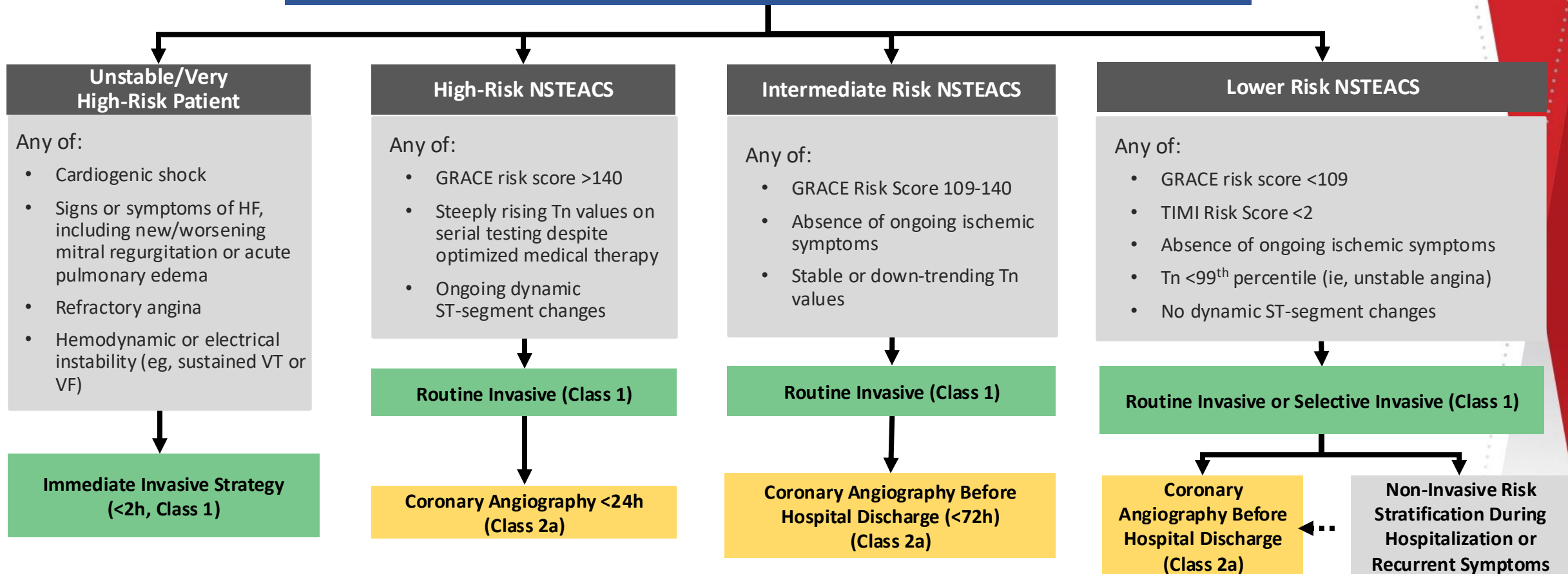


****Clinical Signs of Failed Reperfusion:**

- Ongoing ischemic symptoms
- Persistent ST-segment elevation (<50% resolution of ST-segment elevation in anterior leads or <70% in inferior leads)
- Hemodynamic or electrical instability

Rationale and Timing for a Routine Invasive or Selective Invasive Approach

Choice and Timing of Management Strategy in NSTEMACS



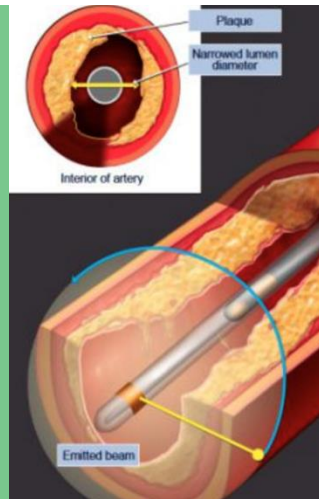
Abbreviations: GRACE indicates Global Registry of Acute Coronary Events; HF, heart failure; NSTEMACS, non-ST-segment elevation acute coronary syndrome; Tn, troponin; TIMI, Thrombolysis in Myocardial Infarction; VF, ventricular fibrillation; and VT, ventricular tachycardia.

Catheterization Lab Considerations in ACS

Radial approach is preferred to a femoral approach to reduce bleeding, vascular complications and mortality (1)



For coronary stent implantation in left main artery or in complex lesions, intracoronary imaging (ICI) with intravascular imaging ultrasound (IVUS) or optical coherence tomography (OCT) is recommended for procedural guidance to reduce ischemic events (1)

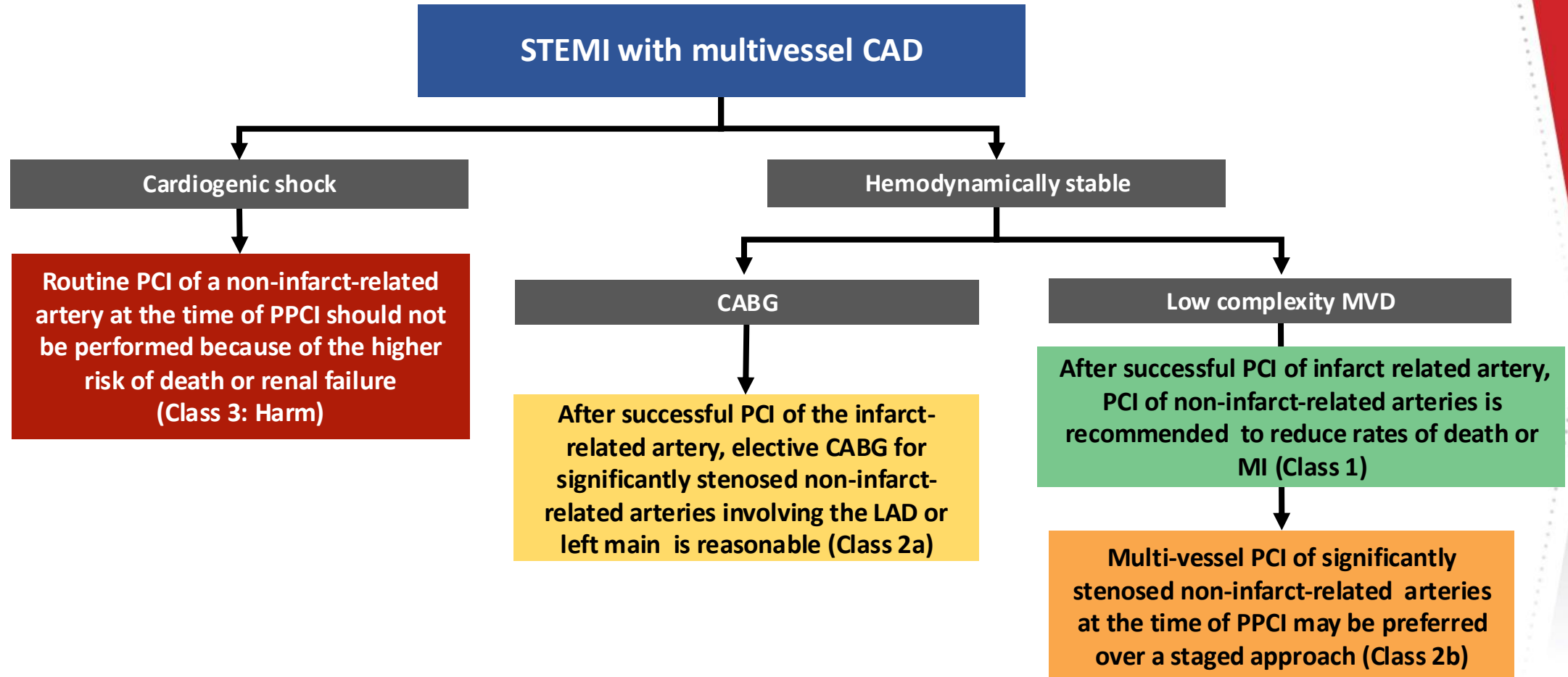


Among patients with STEMI undergoing PPCI, manual aspiration thrombectomy should not be performed routinely (3: No benefit)

Source: Arneja Heart Institute

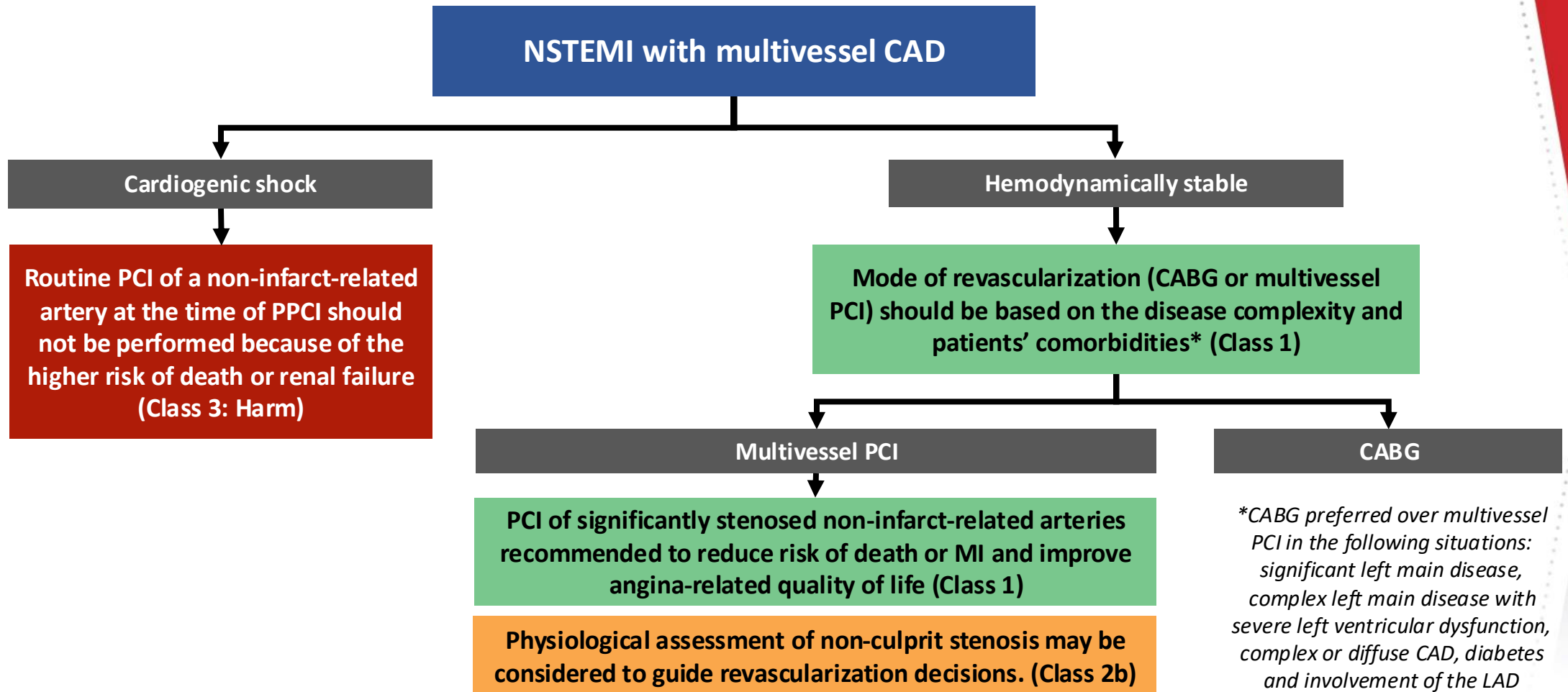
Abbreviations: ACS indicates acute coronary syndrome; STEMI, ST-elevation myocardial infarction; and PPCI, primary percutaneous coronary intervention.

Management of the Non-Infarct-Related Artery in STEMI



Abbreviations: CAD indicates coronary artery disease; CABG, coronary artery bypass graft; STEMI, ST-elevation myocardial infarction; PCI, percutaneous coronary intervention; and PPCI, primary percutaneous coronary intervention.

Management of the Non-Culprit Lesions in NSTEMI-ACS



Abbreviations: CAD indicates coronary artery disease; CABG, coronary artery bypass graft; NSTEMI-ACS, non-ST-elevation acute coronary syndrome; PCI, percutaneous coronary intervention; and PPCI, primary percutaneous coronary intervention.

Revascularization in ACS with Cardiogenic Shock

COR	RECOMMENDATIONS
1	 In patients with ACS and CS or hemodynamic instability, emergency revascularization of the culprit vessel by PCI or with CABG is indicated to improve survival, irrespective of time from symptom onset.
3: HARM	 In patients with ACS complicated by CS, routine PCI of a non-infarct artery at the time of PPCI should not be performed because of the higher risk of death or renal failure.

Electrical Complications and Prevention of Sudden Cardiac Death After ACS

Ventricular Arrhythmias

In patients post MI, implantable cardioverter-defibrillator implantation is recommended in selected patients with an LVEF $\leq 40\%$ at least 40 days post MI and at least 90 days post revascularization to reduce death.*
(Class 1)

Ventricular Arrhythmias

In patients post ACS, ICD implantation is reasonable in patients with clinically relevant ventricular arrhythmias more than 48 hours and within 40 days post MI to improve survival.*
(Class 2a)

Ventricular Arrhythmias

In patients early after MI, usefulness of a temporary wearable cardioverter-defibrillator is uncertain in patients with an LVEF $\leq 35\%$ to improve survival.
(Class 2b)

Bradyarrhythmias

In patients presenting with an acute MI with sustained evidence of second-degree Mobitz type II atrioventricular block, high-grade atrioventricular block, alternating bundle-branch block, or third-degree atrioventricular block (persistent or infranodal), permanent pacing is indicated.†
(Class 1)

**Adapted from 2017 AHA/ACC/HRS Guideline for Management of Patients With Ventricular Arrhythmias and the Prevention of Sudden Cardiac Death*

†Adapted from 2018 ACC/AHA/HRS Guideline on the Evaluation and Management of Patients With Bradycardia and Cardiac Conduction Delay

Abbreviations: ACS indicates acute coronary syndrome; ICD, implantable cardioverter defibrillator; LVEF, left ventricular ejection fraction; and MI, myocardial infarction.

In-Hospital Issues in the Management of ACS

Cardiac Intensive Care Unit



ACS and any of the following:

- Ongoing angina
- Hemodynamic instability
- Uncontrolled arrhythmias
- Suboptimal reperfusion
- Cardiogenic shock

**Admit to CICU
(Class 1)**

Telemetry Monitoring



**In ACS patients,
telemetry monitoring is
recommended to reduce
cardiovascular events
with duration
determined by cardiac
risk.**

(Class 1)

Echocardiogram



**In patients with ACS, an
assessment of LVEF is
recommended prior to
hospital discharge to
guide therapy and for
risk stratification.**

(Class 1)

Blood Transfusions



**In patients with ACS and
acute or chronic anemia,
blood transfusion to achieve
a hemoglobin level ≥ 10 g/dL
may be reasonable to reduce
cardiovascular events.**

(Class 2b)

Patient Education, Lifestyle Modifications, Medication, and Follow-up Care

Hospital admission



Education about CAD, diagnostic tests, procedural results



Return to physical and sexual activity, work and travel

Lifestyle modifications



Smoking cessation



Healthy diet



Regular exercise

Medications



Antithrombotic therapy
Lipid-lowering therapy
Other therapies as appropriate



Annual influenza vaccination

Follow-up care



Follow-up appointments



Cardiology



Cardiac rehabilitation



Additional testing

Symptom management and psychosocial support

Post-discharge Follow-up and Systems of Care Coordination

Clinical Assessment

- Address comorbidities and risk factors
- Assess;
 - *Ongoing ischemic symptoms*
 - *Bleeding risk*
 - *Need for repeat echocardiogram, staged PCI*
 - *Vaccination status like influenza*
- Perform medication reconciliation

Patient/Caregiver Assessment

- Assess patient/caregiver capacity for self care
- Provide verbal and written educational information related to self care
- Use teach-back method to confirm understanding of self-care, medication regimen and adherence

Communication
Patient centered
Share decision-making

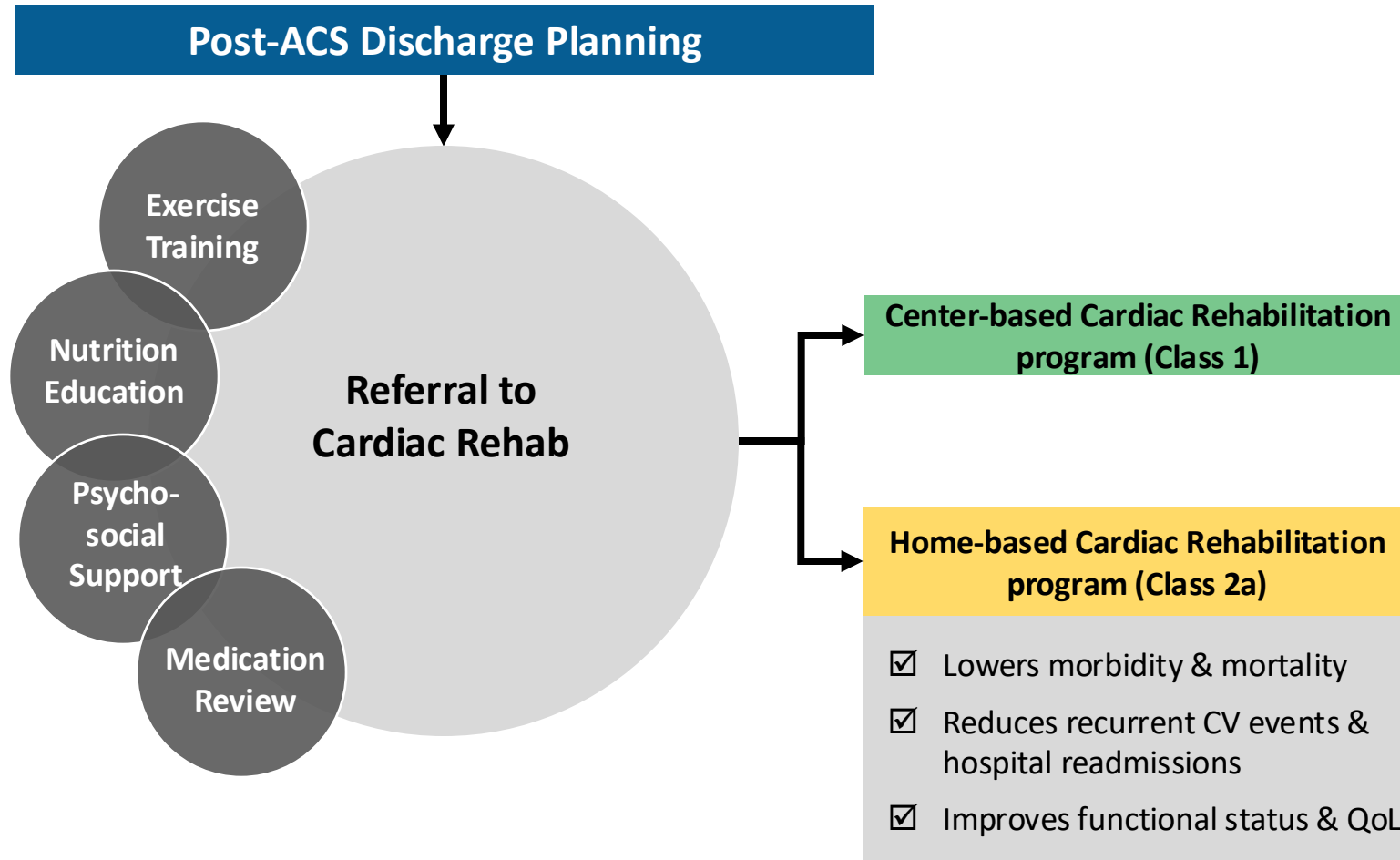
Social Determinants of Health

- Assess and address barriers to obtaining medications
- Refer to pharmacy assistance programs or social worker as appropriate
- Assess and address barriers to attending cardiac rehabilitation

Referrals

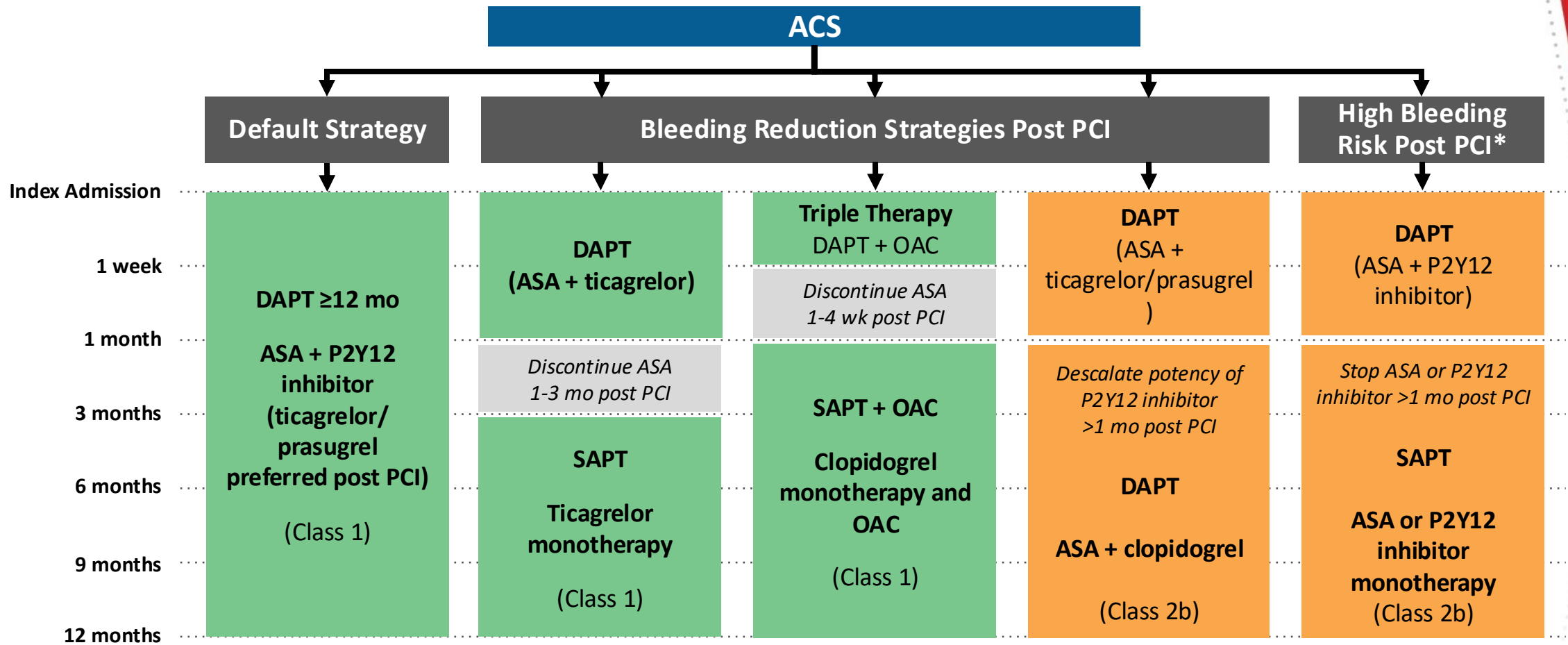
- Confirm referral to cardiac rehabilitation
- Provide educational materials related to cardiac rehabilitation

Cardiac Rehabilitation for Patients Post-ACS



Abbreviations: ACS indicates acute coronary syndrome; CV, cardiovascular; and QoL, quality of life.

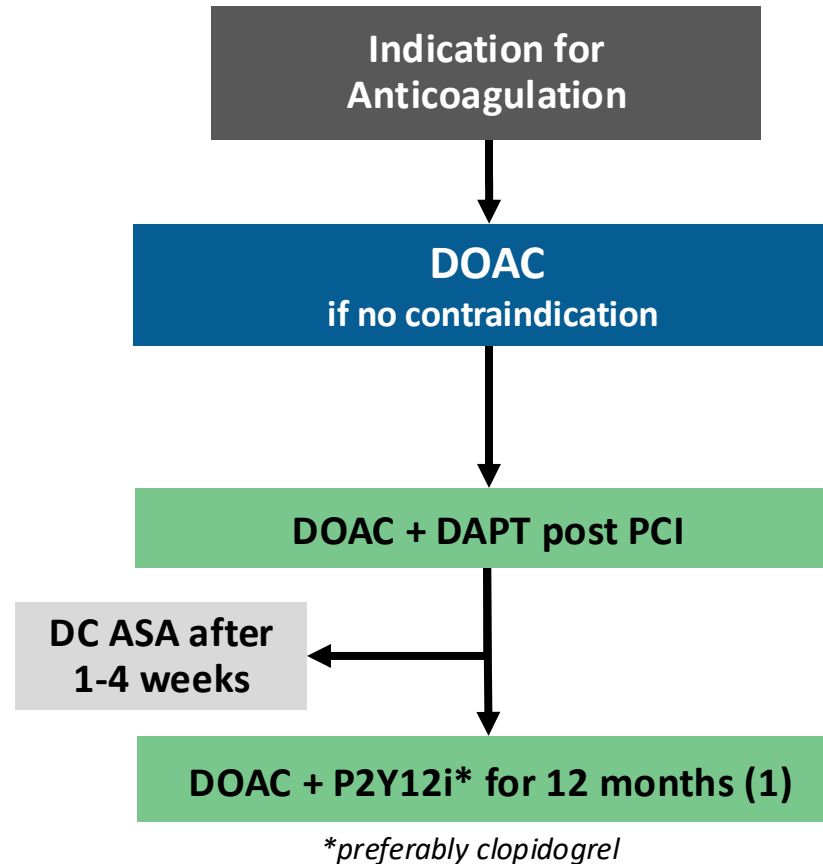
DAPT Strategies in the First 12 Months Post-Discharge



*High bleeding risk discussed in supportive text 5 and outlined in Table 22.

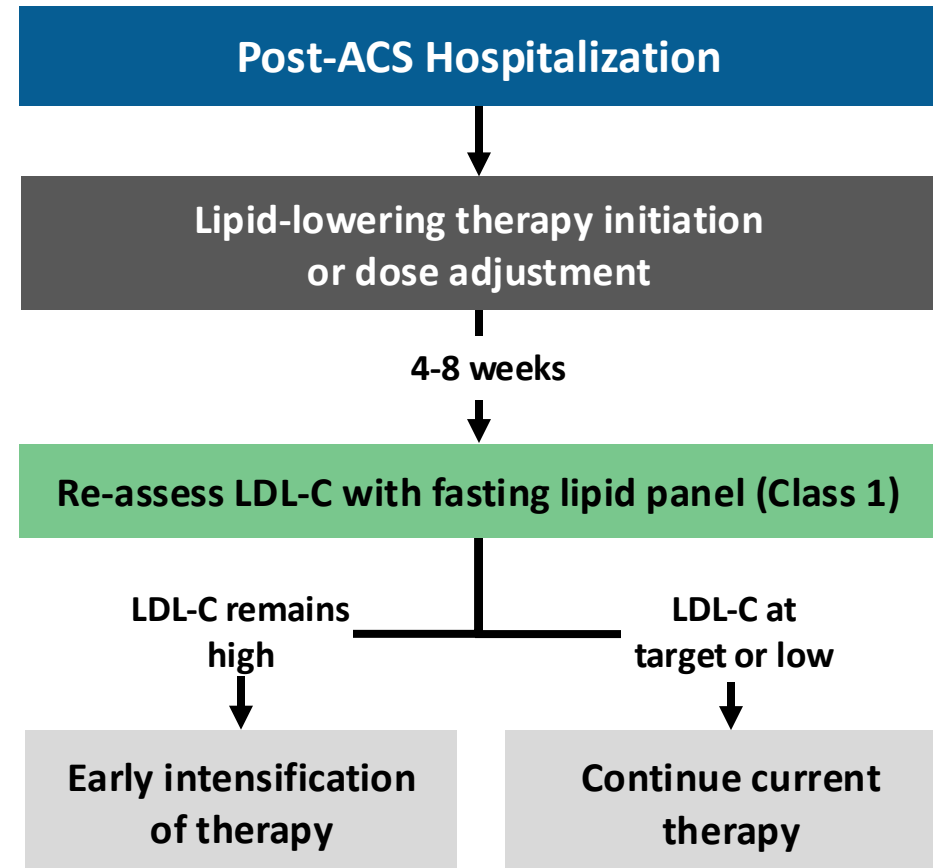
Abbreviations: ACS indicates acute coronary syndrome; ASA, aspirin; DAPT, dual antiplatelet therapy; OAC, oral anticoagulant; PCI, percutaneous coronary intervention; SAPT, single antiplatelet therapy.

Antiplatelet Therapy in Patients on Anticoagulation Post-Discharge



Abbreviations: AF indicates atrial fibrillation; ACS, acute coronary syndrome; ASA, aspirin; DAPT, dual antiplatelet therapy; DC, discontinue; DOAC, direct-acting oral anticoagulant; PCI, percutaneous coronary intervention; and VTE, venous thromboembolism.

Reassessment of Lipid Levels Post-Discharge



Abbreviations: ACS indicates acute coronary syndrome; and LDL-C, low-density lipoprotein cholesterol.

Immunization Management

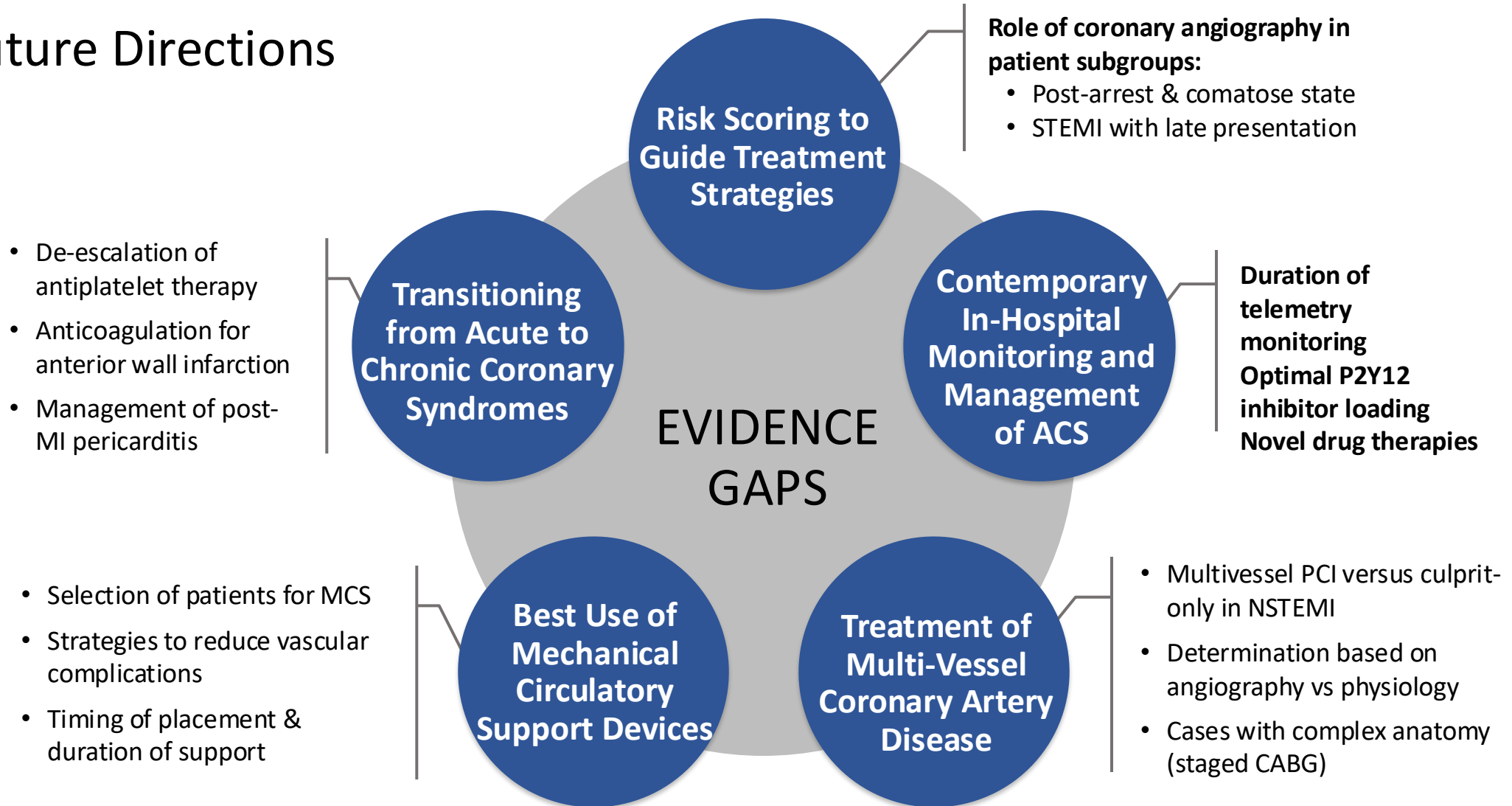


COR	RECOMMENDATIONS
1	In patients with ACS without a contraindication, annual influenza vaccination is recommended to reduce the risk of death and MACE.

Abbreviations: ACS indicates acute coronary syndrome; and MACE, major adverse cardiovascular event.

Rao, S.V., et al. 2025 AHA/ACC/ACEP/NAEMSP/SCAI Guideline for Acute Coronary Syndromes. *Circulation*.

Future Directions



Abbreviations: ACS indicates acute coronary syndromes; CABG, coronary artery bypass grafting; MCS, mechanical circulatory support; MI, myocardial infarction; NSTEMI, non-ST-elevation myocardial infarction; STEMI, ST-elevation myocardial infarction; and PCI, percutaneous coronary intervention.

Acknowledgments

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